

GENE REGULATORY NETWORK INFERENCE FROM INTEGRATIVE MULTI-OMICS DATA

By - Vishak Kashyap K, UG4 CND

Advisor - Dr Vinod PK

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Weekly Update

SC-MO-GRN-DB

Introduction

- **The Core Problem:** Traditional gene regulatory network (GRN) benchmarks primarily rely on bulk sequencing or single-modality transcriptomics (scRNA-seq). They miss critical epigenetic, 3D chromatin, and causal perturbation mechanisms.
- **Benchmarking Fragmentation:** Evaluating GRN inference algorithms currently suffers from unstandardized ground-truth networks, disparate data preprocessing, and inconsistent selection criteria across studies.
- **Unified Platform:** A comprehensive, publicly accessible database (<https://scmogrndb.psu.edu>) unifying high-confidence reference networks with tissue-matched single-cell multiomic datasets.
- **Scale:** 22+ Million experimentally supported regulatory edges, 2.3+ Million harmonized cells across 6 molecular modalities, and 30 Reference Networks alongside 31 Single-Cell Datasets across 16 human/mouse tissues

Reference Network Curation & Evidence Tiers

- **Localization-Based Networks:** Derived from ChIP-seq and ChIP-chip peak bindings mapped to nearest Transcription Start Sites (TSS) to capture physical TF-DNA binding.
- **Perturbation-Based Networks:** Curated from TF knockouts, knockdowns, overexpression, and scCRISPR screens to provide functional/causal evidence of gene expression changes.
- **Dual-Evidence Networks:** Built by intersecting localization and perturbation datasets for the same cell type, requiring simultaneous physical binding and functional readout to substantially reduce false positives.
- **Supplementary Networks:** Incorporates text-mined (TRRUST), literature-curated (expert Boolean models), and Protein-Protein Interaction (PPI via STRING) networks for broader exploratory scope.
- **Network Statistics:** Houses 30 Reference Networks (18 Human, 12 Mouse), spanning 1,616 Transcription Factors, 63,512 Target Genes, and 22,389,021 Total Edges.

Single-Cell Multiomic Datasets & Curation Pipeline

- **Filtering Criteria:** Strictly isolates wild-type or untreated control cells to ensure baseline cellular states match reference network contexts.
- **Cell-Type Stratification:** Multicellular datasets are separated into homogeneous cell-type subsets prior to ingestion.
- **Quality Control:** Low-quality cells are removed based on standard single-cell metrics to minimize technical noise.
- **Multiomic Modality Distribution:** Covers 31 Datasets / 2.32M Cells across scRNA-seq, scATAC-seq, scChIP-seq, scDNA-Met, scHi-C, and scCRISPR-seq.
- **Joint Assays:** Includes 7 joint datasets featuring simultaneous multiomic measurements (e.g., paired scRNA-seq + scATAC-seq within individual cells).

Web Platform Architecture & Analytical Tools

- **Browsing Features:** Allows multi-parameter filtering by Organism (Human/Mouse), Tissue/Cell Type (e.g., ESC, HSC, K562, GM12878), Data Modality, and Evidence Type.
- **Targeted Search:** Queries specific TFs, TGs, or TF-TG pairs while dynamically displaying the backing evidence (Localization, Perturbation, Dual, etc.) and direct PMID references.
- **GRN Performance Evaluation:** Users upload custom/inferred GRNs to calculate precision, recall, F1 score, Jaccard index, and AUPRC against chosen reference networks.
- **Hypergeometric Pathway Enrichment:** Evaluates overrepresented pathways for TF target sets against background expressed genes.
- **Graph Topology Controls:** Computes centrality metrics (Degree, Betweenness, Closeness, PageRank) and quantifies overlap across shared TF regulons.

Summary, Limitations, & Future Directions

- **Key Advantages:** Eliminates preprocessing overhead for researchers by providing standardized, tissue-matched inputs and ground truths.
- **Enhanced Evaluation:** Expands ground-truth evaluation beyond standard ChIP-seq binding to direct functional causality (CRISPR/perturbations).
- **Cell-Type Bias Limitation:** Strong representation in cell lines (e.g., K562, ESC, HSC), while rare cell populations or disease states remain underrepresented.
- **Reference Sparsity Limitation:** Dual-evidence networks offer higher confidence but significantly reduced regulon size due to limited overlapping experimental assays.
- **Future Outlook:** Functions as a living resource with continuous integration of newly published paired joint multi-omic assays (e.g., single-cell multiome) to avoid integrative batch effects.
- **Community Standard:** Designed to act as an open "UCI Machine Learning Repository" equivalent for the single-cell systems biology community.

EpiAwareNet

Published in ACM KDD 2026 Conference

Datasets Used & its size

Dataset	Species	Cells (N)	Genes (G)	Peaks (P)	Edges (E)	Regulators (T)
mN (tomato)	Tomato	8,748	34,074	53,570	170,300	2,381
pN (tomato)	Tomato	12,448	34,074	55,930	170,300	2,381
PBMC	Human	12,012	36,600	111,856	5,092	118
Mouse brain	Mouse	4,881	32,284	42,755	5,064	819

- **Tomato datasets:** root multiome under normal (pN) vs. low-nitrogen (mN) conditions; ground truth from dataset-specific bulk regulator–target links
- **PBMC:** 10x Genomics human PBMC multiome; ground truth from DoRothEA
- **Mouse brain:** 10x Genomics embryonic mouse brain multiome; ground truth from TRRUST
- Regulator–target edges split 80% train / 20% validation; only train edges used for supervision, metrics reported on held-out validation edges

Data Preprocessing

- **RNA**: library-size normalization followed by $\log_1 p$ transform ($\log(1+\text{count})$)
- **ATAC**: dataset-specific normalization with log-transform applied in the dataloader
- **Gene-peak candidate sets** (C_g): for each gene, candidate peaks selected within a $\pm 5\text{kb}$ window around the gene, ranked by genomic distance, and truncated to a fixed maximum width (capped at 50 peaks/gene) for batching which avoids fixed/hard-coded peak-to-gene mapping
- **Bulk-derived priors**: regulator–target links mapped into the model's gene vocabulary; matched edges used as noisy positive labels in Stage 2; all other pairs treated as unlabeled
- QC applied to remove low-quality cells and low-coverage genes/peaks
- Barcode alignment performed across paired RNA/ATAC modalities (already paired by construction in multiome assays)

Model Architecture - Stage 1

Multi-omic Transformer backbone (f_θ); learns joint gene–peak representations per cell

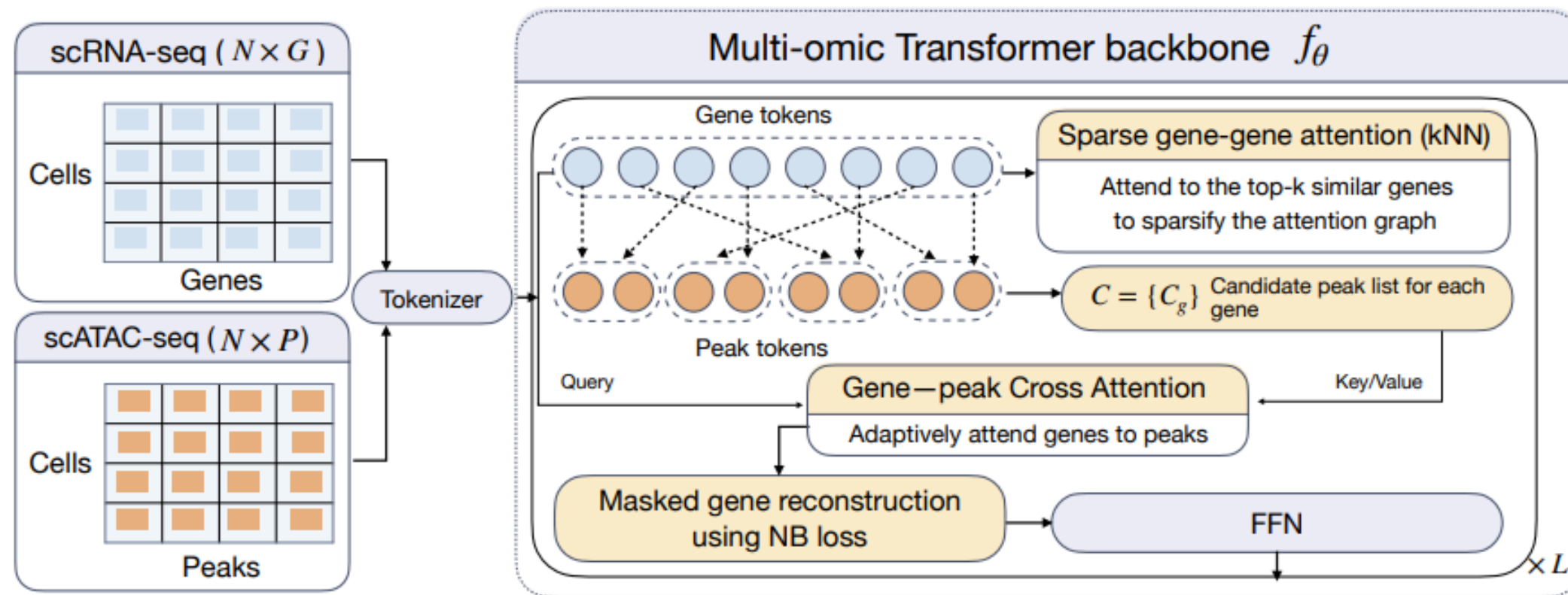
- 6 Transformer layers, hidden dim = 256, 8 attention heads, dropout 0.1
- **Sparse gene–gene self-attention:** each gene attends only to its k-nearest-neighbor genes (cosine similarity in latent space) - reduces cost from $O(G^2)$ to $O(G \cdot k_{NN})$
- **Gene-peak candidate-constrained cross-attention:** each gene attends only to peaks in its precomputed candidate set C_g (optional Top-K routing, default $K=8$), a data-driven, gene-specific aggregation of accessibility signal instead of fixed peak-to-gene rules
- Peak tokens updated via lightweight feed-forward layer (no back-attention to genes)
- **Pretraining objective:** masked gene expression reconstruction (15% genes masked per cell) using a Negative Binomial (NB) loss to handle overdispersed scRNA-seq counts

Model Architecture - Stage 2

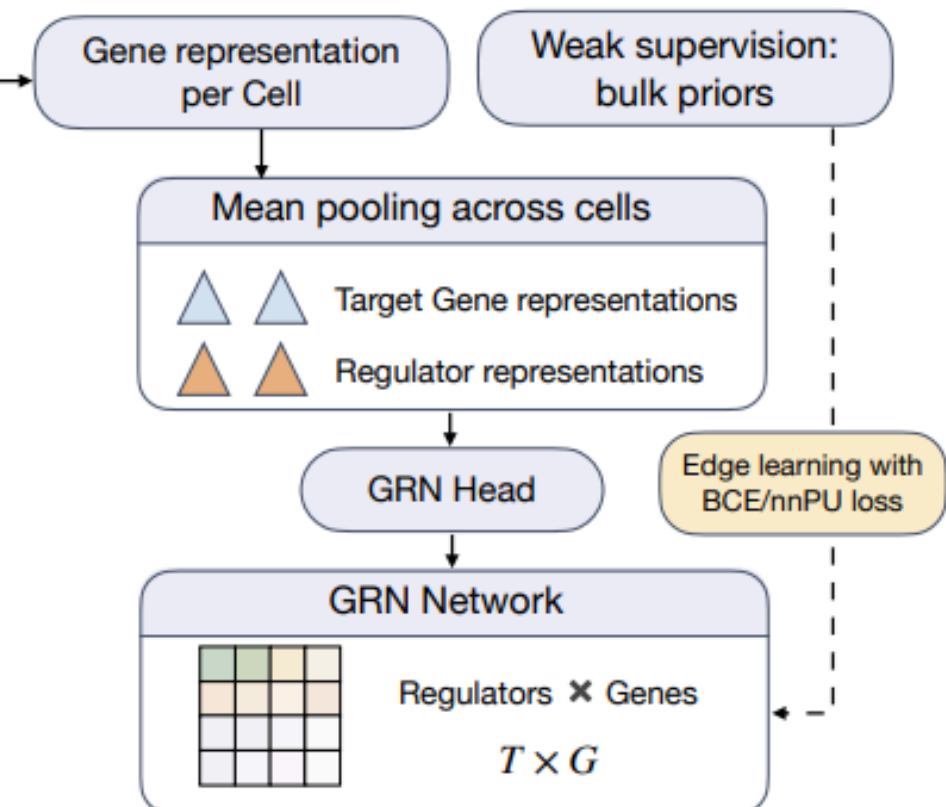
- Backbone f_θ is frozen; a lightweight MLP prediction head (h_ϕ) is trained on top
- Gene/regulator embeddings pooled (averaged) across cells $\rightarrow$ context-level representations
- Prediction head scores each regulator–target pair via concatenated embeddings $\rightarrow$ sigmoid $\rightarrow$ confidence score (probability of regulatory edge)
- **Weak supervision setup:** bulk-derived GRN treated as noisy positive-unlabeled (PU) problem
 - EpiAwareNet (BCE): treats sampled unlabeled pairs as negatives; optimizes for global ranking quality
 - EpiAwareNet-nnPU: non-negative PU objective; more conservative, concentrates confidence on fewer top edges — better for early/precision-focused retrieval
- **Output:** weighted adjacency matrix $A \in [0,1]^{(T \times G)}$ representing regulator $\rightarrow$ gene confidence scores

Model Architecture Diagram

A Stage 1: Representation Learning



B Stage 2: GRN Fine-Tuning



Results

Method	AUPRC	AUROC	P@100	AUPRC Ratio
mN (tomato)				
WGCNA	0.03349	0.5080	0.11	1.0665
GRNBoost2	0.04505	0.6534	0.00	1.4347
scGPT+head	0.03139	0.4998	0.01	0.9999
scGLUE+head	0.03312	0.5055	0.17	1.0550
SCENIC+	0.03160	0.5008	0.13	1.0062
Pando	0.03160	0.5009	0.11	1.0076
RNA-only	0.05238	0.6550	0.16	1.6683
Gene-peak rand.	0.06561	0.6964	0.17	2.0897
EPIAWARENET	0.08458	0.7415	0.13	2.6939
EPIAWARENET-NNPU	0.07081	0.6862	0.25	2.2553
pN (tomato)				
WGCNA	0.03307	0.5055	0.23	1.0532
GRNBoost2	0.04435	0.6435	0.00	1.4126
scGPT+head	0.03139	0.4998	0.01	0.9999
scGLUE+head	0.03173	0.5016	0.07	1.0105
SCENIC+	0.03180	0.5019	0.09	1.0134
Pando	0.03190	0.5013	0.17	1.0144
RNA-only	0.08173	0.7106	0.24	2.6058
Gene-peak rand.	0.06917	0.6910	0.23	2.2053
EPIAWARENET	0.07451	0.7340	0.16	2.3756
EPIAWARENET-NNPU	0.07090	0.6958	0.26	2.2605

Method	AUPRC	AUROC	P@100	AUPRC Ratio
PBMC (human)				
WGCNA	0.00040	0.5151	0.00	1.3174
GRNBoost2	0.00041	0.5713	0.00	1.3677
scGPT+head	0.00046	0.5059	0.01	1.5338
scGLUE+head	0.00034	0.5147	0.00	1.1371
SCENIC+	0.00036	0.5059	0.01 [*]	1.1950
Pando	0.00064	0.5071	0.03	2.1251
RNA-only	0.00165	0.7782	0.00	5.4754
Gene-peak rand.	0.00150	0.7540	0.00	4.9777
EPIAWARENET	0.00244	0.8218	0.03	8.0970
EPIAWARENET-NNPU	0.00135	0.7459	0.01	4.4799
Mouse brain				
WGCNA	0.00011	0.5074	0.00	1.2497
GRNBoost2	0.00010	0.5295	0.00	1.1048
scGPT+head	0.00015	0.5936	0.01	1.7334
scGLUE+head	0.00011	0.5302	0.00	1.2245
SCENIC+	0.00009	0.5020	0.00	1.0188
Pando	0.00010	0.5018	0.00	1.0874
RNA-only	0.00009	0.5205	0.00	1.0191
Gene-peak rand.	0.00014	0.5715	0.00	1.5853
EPIAWARENET	0.00057	0.7263	0.01	6.4543
EPIAWARENET-NNPU	0.00027	0.5170	0.01	3.0573

Limitations

- Candidate peak sets ($\pm 5\text{kb}$) may miss distal regulatory elements (enhancers)
- Bulk-derived priors may carry biases toward well-studied pathways/species and may not reflect condition-specific single-cell regulation
- Evaluation relies on incomplete gold-standard references (possible false negatives); global metrics may understate true performance
- Framework evaluates generalization to held-out edges under a prior-informed setting, not fully prior-free de novo discovery

Future Directions

- Enrich candidate construction to better capture distal regulatory elements
- Extend inference to context-specific GRNs across different cell types/conditions while preserving computational efficiency
- Further explore trade-offs between global ranking (BCE) and high-confidence early retrieval (nnPU) objectives



THANK YOU!

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